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Portfolio Risk-Return Analysis Using Python

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INDEX

Section	Title
1	Executive Summary
2	Introduction
3	Objectives of the Analysis
4	Dataset and Methodology
5	Individual Stock Risk-Return Analysis
6	Risk-Return Trade-Off
7	Visualization Analysis
8	Portfolio Construction
9	Portfolio A vs Portfolio B
10	Excel vs Python Validation

11	Higher-Order Analysis
12	Final Investment Recommendation
13	Limitations of the Analysis
14	Conclusion

List of Figures

Figure No.	Figure Title
Figure 1	Risk vs Return of Selected Stocks
Figure 2	Portfolio Risk vs Return Comparison
Figure 3	Sharpe Ratio Comparison
Figure 4	Beta Comparison

List of Tables

Table No.	Table Title
Table 1	Individual Stock Risk-Return Metrics
Table 2	Portfolio A – Equal-Weighted Allocation
Table 3	Portfolio B – Student-Designed Allocation
Table 4	Portfolio A vs Portfolio B Comparison
Table 5	Excel vs Python Validation
Table 6	₹10,00,000 Investment Allocation

1. Executive Summary

This report analyses the risk-return performance of three Indian listed companies — HDFC Bank, TCS and Reliance Industries — against the NIFTY 50 benchmark, using sixty months of closing price data from January 2021 to December 2025. The analysis was first performed in Microsoft Excel and then replicated in Python to validate the calculations. A risk-free rate of 6% per annum and a market-return assumption of 15% per annum, confirmed in both the workbook and the notebook, are used throughout for CAPM, Sharpe Ratio and Treynor Ratio computations.

Reliance Industries delivered the highest annualized return (13.49% in Python; 11.07% in Excel) and the highest beta (1.1548). HDFC Bank produced a moderate return (9.01%/8.66%) with the lowest risk (16.97%) and a below-market beta (0.8939). TCS was the weakest performer, with the lowest return (2.87%/2.78%) and a negative Sharpe Ratio. On a risk-adjusted basis, Reliance Industries was the strongest performer, with the highest Sharpe Ratio (0.3711) and Treynor Ratio (6.49%).

Two Python-based portfolios were evaluated. Portfolio A (equal-weighted, 33.33% each) generated an expected return of 8.38%, risk of 14.66% and a Sharpe Ratio of 0.1622. Portfolio B (40% HDFC Bank, 10% TCS, 50% Reliance Industries) produced a higher return of 10.59% for only slightly higher risk (15.33%), giving a superior Sharpe Ratio of 0.2995, largely because it reduces exposure to the underperforming TCS position.

The Excel-versus-Python validation confirms that average return, standard deviation, beta and correlation are effectively identical across platforms. Both platforms use the same 15% NIFTY 50 market-return assumption for CAPM, so the modest differences observed in CAPM, Sharpe and Treynor figures for HDFC Bank and TCS are attributable to formula and reference implementation differences within Excel rather than to a differing market-return input; for Reliance Industries, Excel and Python produce an identical CAPM value of 16.39%.

Based on this evidence, Portfolio B is recommended for a moderately risk-averse investor over a five-year horizon. The ₹10,00,000 investment is accordingly allocated as ₹4,00,000 (40%) to HDFC Bank, ₹1,00,000 (10%) to TCS and ₹5,00,000 (50%) to Reliance Industries, balancing HDFC Bank's stability with Reliance Industries' superior risk-adjusted return while retaining a small diversifying position in TCS.

2. Introduction

Every investment decision involves a trade-off between expected return and the risk taken to earn it. Evaluating a stock purely on historical return can be misleading, since a high return accompanied by disproportionately high volatility may leave an investor worse off, in risk-adjusted terms, than a stock with a more modest but stable return. This report examines HDFC Bank, TCS and Reliance Industries against the NIFTY 50 benchmark to determine not merely which stock produced the highest return, but which offered the best compensation for the risk undertaken.

The Sharpe Ratio and Treynor Ratio are used because they normalize return by risk, allowing a fair comparison between securities of differing volatility and market sensitivity (beta). CAPM is used as a complementary benchmark, indicating the return an asset should theoretically deliver given its systematic risk, compared against its actual return.

Diversification is examined through two portfolios: an equal-weighted Portfolio A, and a student-designed Portfolio B that overweights the stronger risk-adjusted performer, illustrating the benefit of directing capital toward securities with superior risk-adjusted characteristics rather than spreading capital uniformly.

The analysis was performed independently in Excel and Python, both to validate that the Excel formulas were implemented correctly and to demonstrate the same analysis performed programmatically. Discrepancies between the two platforms are documented and explained rather than reconciled artificially.

3. Objectives of the Analysis

The specific objectives of this analysis are to:

- Calculate monthly returns for HDFC Bank, TCS and Reliance Industries.
- Calculate monthly returns for the NIFTY 50 benchmark index.
- Measure the average monthly return for each stock.
- Compute the annualized return for each stock and for the benchmark.
- Measure risk through the annualized standard deviation of returns.
- Calculate beta as a measure of systematic (market) risk for each stock.
- Calculate the correlation of each stock's returns with NIFTY 50 returns.
- Compute the CAPM required rate of return for each stock.
- Calculate the Sharpe Ratio for each stock and portfolio.
- Calculate the Treynor Ratio for each stock and portfolio.
- Construct an equal-weighted Portfolio A and a student-designed Portfolio B.
- Compare Excel-based and Python-based results and explain any differences.
- Evaluate risk-adjusted performance across stocks and portfolios.
- Recommend a ₹10,00,000 investment allocation for a five-year horizon.

4. Dataset and Methodology

4.1 Dataset

The analysis uses monthly closing prices for HDFC Bank, TCS and Reliance Industries, together with the NIFTY 50 index level, covering sixty months from January 2021 to December 2025 — a five-year period consistent with the assignment's three-to-five-year requirement. Both the Excel workbook and the Python notebook use the same monthly closing price series for all four instruments, so the return frequency is applied consistently throughout, as required by the assignment brief.

A risk-free rate of 6% per annum and a NIFTY 50 market-return assumption of 15% per annum are used consistently across both platforms. The 6% risk-free rate and 15% market return are confirmed in the 'RF' and 'RM' cells of the Excel workbook (HDFC, TCS, Reliance sheets); in the Python notebook, the risk-free rate is set directly at 6% while the NIFTY 50 annualized return is computed from the actual price data, arriving at an equivalent value of approximately 15%. Because both platforms therefore rest on the same market-return assumption, any residual differences in CAPM-derived figures (Section 10) reflect formula or reference implementation differences within Excel rather than a differing input assumption.

4.2 Methodology and Formulas

Monthly Return: $(\text{Current Month Close} \div \text{Previous Month Close}) - 1$, calculated on month-end closing prices; the first month is excluded as it has no prior price.

Average Monthly Return: Arithmetic mean of the monthly return series.

Annualized Return: $(1 + \text{Average Monthly Return})^{12} - 1$, applied identically in both platforms; small differences in the input average return cause small differences in the annualized figure (Section 10).

Annualized Risk: $\text{Monthly Standard Deviation} \times \sqrt{12}$, applied identically in both platforms.

Beta: $\text{Covariance}(\text{Stock Return}, \text{NIFTY 50 Return}) \div \text{Variance}(\text{NIFTY 50 Return})$.

Correlation: Pearson correlation coefficient between each stock's and NIFTY 50's monthly returns.

CAPM Required Return: $\text{Risk-Free Rate} + \text{Beta} \times (\text{Market Return} - \text{Risk-Free Rate})$, with Risk-Free Rate = 6% and Market Return = 15% used consistently in both Excel and Python. Small differences in the resulting CAPM figures (Section 10) arise from formula/reference implementation rather than from differing input assumptions.

Sharpe Ratio: $(\text{Annualized Return} - \text{Risk-Free Rate}) \div \text{Annualized Standard Deviation}$ — excess return per unit of total risk.

Treynor Ratio: $(\text{Annualized Return} - \text{Risk-Free Rate}) \div \text{Beta}$ — excess return per unit of systematic risk.

The Python notebook implements these formulas on the pandas return series using NumPy/pandas statistical functions (mean, std, cov, var, corr); the Excel workbook implements the equivalent calculations using native worksheet functions on the same monthly price data.

5. Individual Stock Risk-Return Analysis

Table 1 summarizes the Python-computed risk-return metrics for the three selected stocks and the NIFTY 50 benchmark. Python figures are used as the primary reference in this table because the Python notebook computes the NIFTY 50 annualized return directly from the data, allowing the benchmark row to be populated on a consistent basis; the corresponding Excel-based figures are presented separately in the Excel vs Python validation table in Section 10.

Metric	HDFC Bank	TCS	Reliance Industries	NIFTY 50
Average Monthly Return	0.72%	0.24%	1.06%	—
Annualized Return	9.01%	2.87%	13.49%	15.00%
Annualized Std. Deviation (Risk)	16.97%	21.18%	20.19%	12.44%
Beta	0.8939	0.8161	1.1548	1.0000
Correlation with NIFTY 50	0.6552	0.4794	0.7114	1.0000
CAPM Required Return	14.04%	13.34%	16.39%	—
Sharpe Ratio	0.1775	-0.1479	0.3711	—
Treynor Ratio	3.37%	-3.84%	6.49%	—

5.1 HDFC Bank

HDFC Bank delivered an annualized return of 9.01% with the lowest risk among the three stocks (16.97%) and a beta of 0.8939, indicating lower volatility than the broader market. Its correlation with NIFTY 50 (0.6552) is the second-highest of the three. Its CAPM required return of 14.04% exceeds its actual return, implying underperformance against its theoretical risk-based expectation; nonetheless, its Sharpe Ratio (0.1775) and Treynor Ratio (3.37%) are both positive. Overall, HDFC Bank behaves as a defensive, lower-volatility holding within the selected set.

5.2 TCS

TCS was the weakest performer, with an annualized return of only 2.87% — below both the risk-free rate of 6% and its CAPM required return of 13.34%. Its beta (0.8161) and correlation with NIFTY 50 (0.4794) are the lowest of the three, suggesting its price movements were more idiosyncratic. Because its return fell below the risk-free rate, TCS records a negative Sharpe Ratio (-0.1479) and Treynor Ratio (-3.84%). Its high risk (21.18%) combined with its low return makes it the least attractive stock over the period.

5.3 Reliance Industries

Reliance Industries produced the highest annualized return (13.49%) and beta (1.1548), the strongest correlation with the benchmark (0.7114), the highest risk (20.19%), and comfortably the best Sharpe Ratio (0.3711) and Treynor Ratio (6.49%). Its CAPM required return (16.39%, Python) exceeds its realized return,

meaning it technically underperformed CAPM's theoretical requirement — yet it still delivered the best risk-adjusted outcome, since CAPM and Sharpe/Treynor answer different questions (see Section 11, Q1).

5.4 Cross-Stock Comparison

Ranking by annualized return places Reliance Industries first, HDFC Bank second and TCS last; the same ranking holds under Sharpe and Treynor Ratios, so the highest-return stock is also the best risk-adjusted performer here — though this is not a universal rule. TCS demonstrates that a stock can carry the highest risk (21.18%) while failing to generate a commensurate return, exactly what risk-adjusted measures are designed to expose.

6. Risk-Return Trade-Off

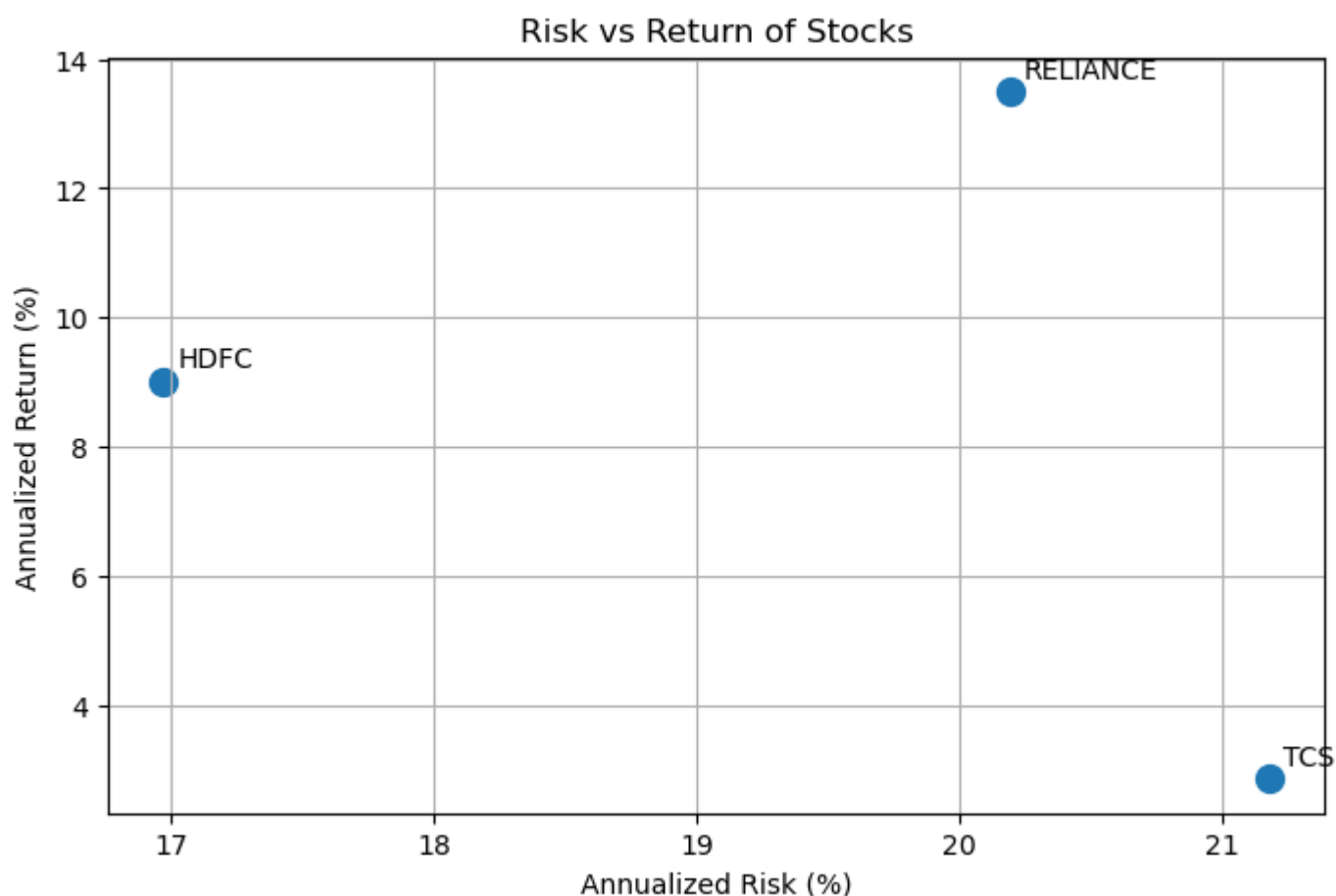
The risk-return relationship is not strictly proportional across the three stocks. Reliance Industries carries the highest risk (20.19%) and the highest return (13.49%), consistent with the expectation that risk should be compensated. TCS, however, carries the highest risk of all three (21.18%) yet produces the lowest return (2.87%) — a clear violation of the risk-return trade-off, and the clearest evidence here that risk alone does not guarantee reward.

HDFC Bank has the lowest risk (16.97%) and lowest beta (0.8939), confirming its role as the most defensive stock. Reliance Industries has the highest beta (1.1548) and strongest correlation with NIFTY 50 (0.7114), making it the most market-sensitive stock, while TCS has both the lowest beta (0.8161) and weakest correlation (0.4794), suggesting its returns are driven more by company- or sector-specific factors than by the broader market.

Reliance Industries, the highest-return stock, is also the highest-risk stock, so it is not automatically the best investment on return alone. However, because its Sharpe and Treynor Ratios remain the highest despite this elevated risk, the additional risk taken is more than compensated for — which is why it is identified as the strongest risk-adjusted performer (Section 5.4; Section 11, Q2), not merely the stock with the largest headline return.

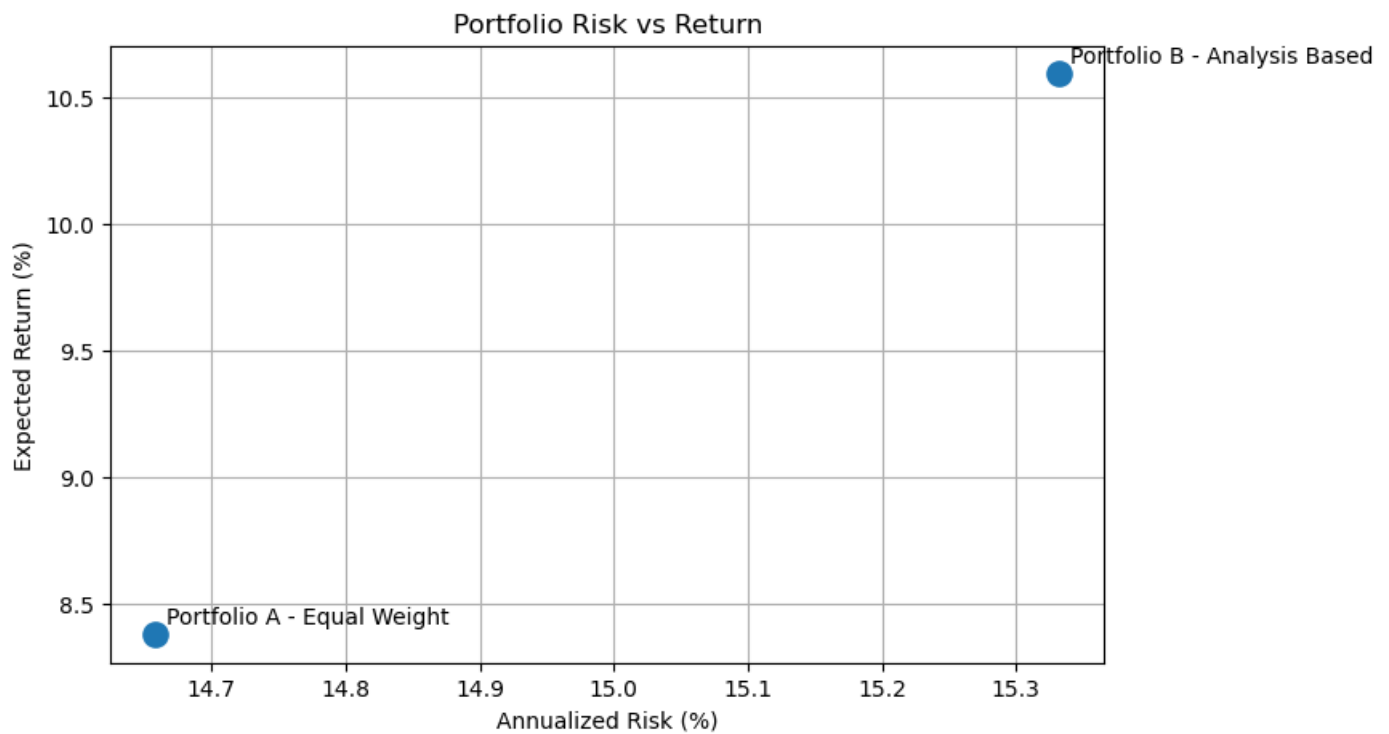
7. Visualization Analysis

The following four visualizations, generated in the accompanying Python notebook, support the quantitative findings above. Each graph is to be inserted manually at the marked location; a written interpretation grounded in the actual computed values follows each placeholder.



Interpretation:

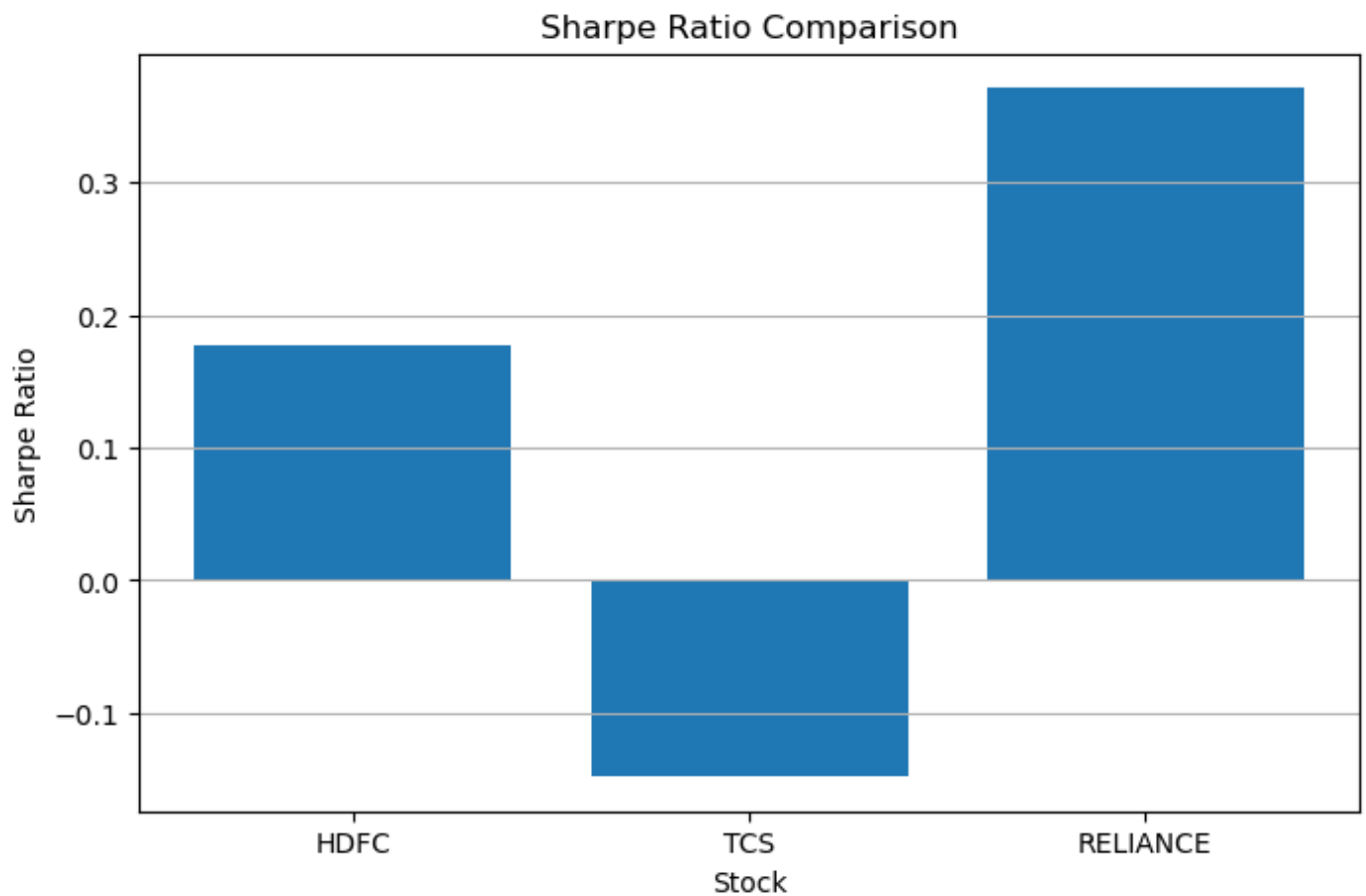
The scatter plot positions Reliance Industries upper-right (highest risk ~20.19%, highest return ~13.49%), TCS upper-middle (high risk ~21.18% but lowest return ~2.87%), and HDFC Bank lower-left (lowest risk ~16.97%, moderate return ~9.01%). This confirms TCS is dominated by the other two stocks — similar or lower risk was available elsewhere for a much higher return — while Reliance Industries occupies the most favourable risk-return position.



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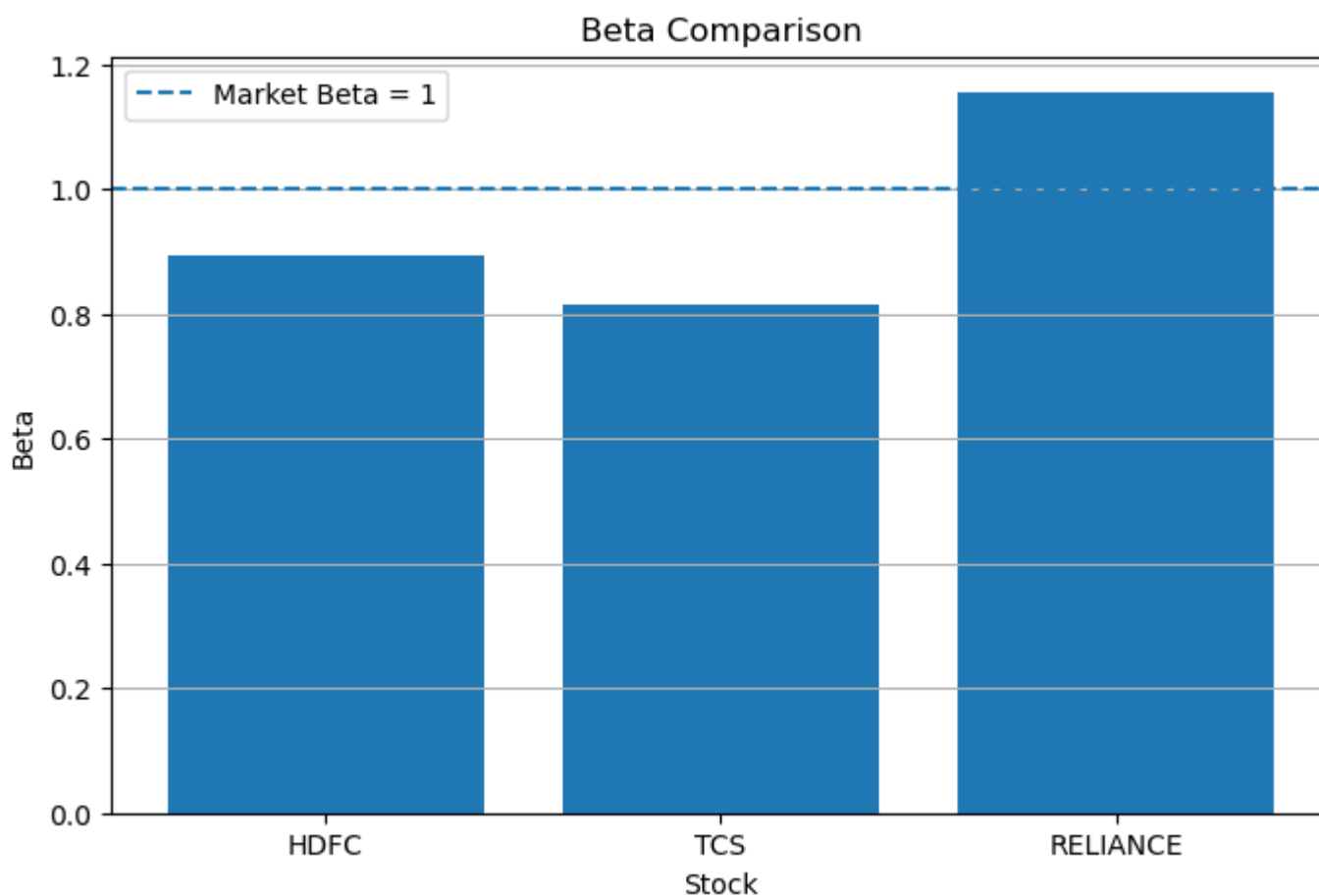
Interpretation:

Portfolio B sits above and only marginally to the right of Portfolio A: its return (10.59%) is noticeably higher than Portfolio A's (8.38%), while its risk (15.33%) is only slightly higher (14.66%). This confirms Portfolio B achieves a materially better return for nearly identical risk, the basis for its superior Sharpe Ratio (0.2995 vs 0.1622) discussed in Section 9.



Interpretation:

Reliance Industries has the tallest bar (0.3711), HDFC Bank a smaller positive bar (0.1775), and TCS a bar below zero (-0.1479). This shows Reliance Industries offered the best compensation for risk taken, HDFC Bank modest compensation, and TCS actually destroyed risk-adjusted value relative to a risk-free investment.



Interpretation:

With a reference line at market beta 1.0, Reliance Industries is the only stock above it (1.1548), confirming greater volatility than the market. HDFC Bank (0.8939) and TCS (0.8161) sit below the line, theoretically less volatile than NIFTY 50 — though, as shown in Section 6, lower beta did not protect TCS from a poor risk-adjusted return.

8. Portfolio Construction

8.1 Portfolio A — Equal-Weighted

Portfolio A allocates capital equally across the three selected stocks, representing a naïve diversification strategy with no active view on relative risk-adjusted performance.

Stock	Weight	Annualized Return (Stock)
HDFC Bank	33.33%	9.01%
TCS	33.33%	2.87%
Reliance Industries	33.33%	13.49%

Expected Return: 8.38% per annum.

Portfolio Risk: 14.66% (annualized standard deviation of the equal-weighted monthly return series).

Portfolio Beta: 0.9550, marginally below the market beta of 1.00, since HDFC Bank's and TCS's below-market betas are only partly offset by Reliance Industries' above-market beta.

Sharpe Ratio: 0.1622.

Portfolio A benefits from diversification in that its risk (14.66%) is lower than the simple average of the three stock risks (19.45%), since the stocks are not perfectly correlated. However, because TCS — the weakest performer — receives the same weight as the two stronger stocks, Portfolio A's return and Sharpe Ratio are held back relative to a more selective allocation.

8.2 Portfolio B — Student-Designed

Portfolio B uses weights of 40% HDFC Bank, 10% TCS and 50% Reliance Industries, as recorded in both the Python notebook and the ₹10 lakh allocation exercise.

Stock	Weight	Annualized Return (Stock)
HDFC Bank	40%	9.01%
TCS	10%	2.87%
Reliance Industries	50%	13.49%

The weighting deliberately minimizes exposure to TCS (10%), the only stock with negative Sharpe and Treynor Ratios, while maximizing exposure to Reliance Industries (50%), the stock with the highest return, Sharpe Ratio (0.3711) and Treynor Ratio (6.49%). HDFC Bank receives an intermediate weight (40%), reflecting its role as the most defensive of the stronger two performers (lowest risk, 16.97%), offsetting some volatility from the larger Reliance Industries position.

Expected Return: 10.59% p.a. — 2.21 pp higher than Portfolio A.

Portfolio Risk: 15.33% — only 0.67 pp higher than Portfolio A's 14.66%.

Portfolio Beta: 1.0166, marginally above the market beta of 1.00.

Sharpe Ratio: 0.2995, materially higher than Portfolio A's 0.1622.

Portfolio B's weighting is quantitatively justified — driven directly by the observed Sharpe Ratios, Treynor Ratios and returns of the three stocks — rather than by a formal optimization procedure. No mean-variance or Sharpe-maximizing routine was applied to derive these weights; accordingly, Portfolio B is described here as a student-designed, evidence-based allocation, not an "optimized" portfolio.

9. Portfolio A vs Portfolio B

Metric	Portfolio A	Portfolio B	Interpretation
Expected Return	8.38%	10.59%	Portfolio B higher by 2.21 pp
Portfolio Risk	14.66%	15.33%	Portfolio B higher by 0.67 pp
Beta	0.9550	1.0166	Portfolio B marginally above market
Sharpe Ratio	0.1622	0.2995	Portfolio B substantially higher

Portfolio B generates a higher return (10.59% vs 8.38%) and carries higher risk (15.33% vs 14.66%) and a marginally higher beta (1.0166 vs 0.9550). Despite the higher risk, its Sharpe Ratio (0.2995 vs 0.1622) is considerably better, meaning the additional return more than compensates for the additional risk — 2.21 pp of extra return for only 0.67 pp of extra risk.

For a moderately risk-averse investor — not risk-seeking, but willing to accept a modest increase in volatility for a meaningfully better risk-adjusted outcome — Portfolio B is the more appropriate choice. Portfolio A would only be preferable to an investor who places overriding importance on minimizing absolute risk, even at the cost of a lower Sharpe Ratio.

10. Excel vs Python Validation

Table below compares the Excel-based and Python-based results for each metric and stock, drawing directly from the 'Excel Vs Python' validation sheet in the workbook and the corresponding notebook outputs.

Metric	Stock	Excel	Python	Difference
Average Monthly Return	HDFC Bank	0.72%	0.72%	0.00 pp
Average Monthly Return	TCS	0.23%	0.24%	0.01 pp
Average Monthly Return	Reliance	0.92%	1.06%	0.14 pp
Annualized Return	HDFC Bank	8.66%	9.01%	0.35 pp
Annualized Return	TCS	2.78%	2.87%	0.09 pp
Annualized Return	Reliance	11.07%	13.49%	2.42 pp
Annualized Std. Deviation	HDFC Bank	16.97%	16.97%	0.00 pp
Annualized Std. Deviation	TCS	21.00%	21.18%	0.18 pp
Annualized Std. Deviation	Reliance	20.36%	20.19%	0.17 pp
Beta	HDFC Bank	0.8939	0.8939	0.0000
Beta	TCS	0.8161	0.8161	0.0000
Beta	Reliance	1.1548	1.1548	0.0000
Correlation	HDFC Bank	0.6552	0.6552	0.0000
Correlation	TCS	0.4794	0.4794	0.0000
Correlation	Reliance	0.7114	0.7114	0.0000
CAPM Required Return	HDFC Bank	14.36%	14.04%	0.32 pp
CAPM Required Return	TCS	13.90%	13.34%	0.56 pp
CAPM Required Return	Reliance	16.39%	16.39%	0.00 pp
Sharpe Ratio	HDFC Bank	0.1568	0.1775	0.0207
Sharpe Ratio	TCS	-0.1497	-0.1479	0.0018
Sharpe Ratio	Reliance	0.3330	0.3711	0.0381
Treynor Ratio	HDFC Bank	2.98%	3.37%	0.39 pp
Treynor Ratio	TCS	-3.88%	-3.84%	0.04 pp

Metric	Stock	Excel	Python	Difference
Treynor Ratio	Reliance	5.82%	6.49%	0.67 pp

Average monthly returns are nearly identical across Excel and Python, with the largest deviation being 0.14 pp for Reliance Industries. Annualized standard deviation, beta and correlation are, for practical purposes, identical (differences of 0.00–0.18 pp, or exactly zero for beta and correlation), strong evidence that the underlying return-series, covariance and variance calculations are implemented consistently and correctly in both platforms.

Annualized return shows a small, consistent upward difference in Python, most visible for Reliance Industries (2.42 pp). Since both platforms apply the same compounding formula, $(1 + \text{Average Monthly Return})^{12} - 1$, this difference stems from small differences in the average monthly return input rather than the annualization formula itself.

The largest and most instructive divergence is in the CAPM required return for HDFC Bank and TCS: HDFC Bank shows 14.36% (Excel) against 14.04% (Python), a gap of 0.32 pp, and TCS shows 13.90% against 13.34%, a gap of 0.56 pp. Both platforms share the same input assumptions — a 6% risk-free rate and a 15% NIFTY 50 market return — so these gaps are not attributable to a differing market-return assumption, but instead reflect formula or cell-reference implementation differences within the Excel workbook's CAPM cells. For Reliance Industries, Excel and Python arrive at an identical CAPM value of 16.39%, confirming that the two platforms agree exactly once the formula references align, and that the HDFC Bank and TCS gaps are an implementation-level artefact rather than a conceptual disagreement over inputs.

Sharpe and Treynor Ratio differences are modest across all three stocks, since these ratios depend primarily on each stock's own annualized return and risk rather than on the market-return assumption. Neither the Excel nor the Python CAPM figures for HDFC Bank and TCS should be regarded as "incorrect": both share the same 6% risk-free rate and 15% market-return assumption, and the residual gap reflects how each platform's CAPM formula is referenced rather than a disagreement over inputs. Overall, the exercise strongly validates the core Excel calculations — particularly beta, correlation and standard deviation — while smaller differences in return-based measures trace to annualization conventions, and the CAPM gaps for HDFC Bank and TCS to formula-reference implementation.

11. Higher-Order Analysis

Q1. Is the stock with the highest return necessarily the most attractive investment?

Reliance Industries has the highest annualized return (13.49%), but return alone does not establish it as the most attractive investment. First, on risk, it also carries the highest standard deviation (20.19%, close to TCS's 21.18%). Second, on the Sharpe Ratio, it still leads (0.3711 vs 0.1775 for HDFC Bank and -0.1479 for TCS), confirming its higher return more than compensates for its higher risk. Third, on the Treynor Ratio, it again leads (6.49% vs 3.37% and -3.84%). Fourth, under CAPM, its required return (16.39%) actually exceeds its realized return, meaning it technically underperformed its theoretical benchmark. The highest-return stock here does also happen to be the best risk-adjusted performer, but this is specific to this dataset — the CAPM comparison, and the counter-example of TCS (highest risk, lowest return), both show that a high return figure alone is not sufficient evidence of an attractive investment.

Q2. Which stock provides the best risk-adjusted performance?

Reliance Industries provides the best risk-adjusted performance. On the Sharpe Ratio, it records 0.3711, above HDFC Bank's 0.1775 and far above TCS's -0.1479. The Treynor Ratio confirms the same ranking: Reliance Industries (6.49%), HDFC Bank (3.37%), TCS (-3.84%), supported by Reliance Industries' highest correlation (0.7114). TCS is unambiguously the weakest, with negative values on both ratios, failing to even match the risk-free return, while HDFC Bank offers modest positive compensation with the lowest absolute risk. Reliance Industries is judged the best risk-adjusted performer.

Q3. Portfolio A vs Portfolio B — recommendation for a moderately risk-averse investor

Portfolio B is recommended over Portfolio A for a moderately risk-averse investor. Its return (10.59%) is 2.21 pp higher than Portfolio A's (8.38%), while its risk (15.33%) is only 0.67 pp higher (14.66%) — a disproportionately small increase in risk relative to the return gained. This is reflected in the Sharpe Ratio, where Portfolio B (0.2995) nearly doubles Portfolio A's (0.1622); beta (1.0166 vs 0.9550) is not different enough to meaningfully change the risk character. Both portfolios hold the same three stocks; the difference lies entirely in the weighting, with Portfolio B reducing exposure to the weak, negative-Sharpe TCS position. Such an investor is better served by Portfolio B.

Q4. ₹10,00,000 investment allocation for a five-year horizon

Stock	Weight	Investment Amount (₹)
HDFC Bank	40%	4,00,000
TCS	10%	1,00,000
Reliance Industries	50%	5,00,000
Total	100%	10,00,000

This allocation follows the Portfolio B weights, shown in Sections 8 and 9 to be the superior risk-adjusted allocation. The largest allocation (50%) goes to Reliance Industries, the stock with the highest return (13.49%), Sharpe Ratio (0.3711) and Treynor Ratio (6.49%). HDFC Bank receives the second-largest

allocation (40%), justified by its role as the most defensive holding (lowest risk, 16.97%, and beta, 0.8939, among the stronger two performers), moderating the volatility introduced by the large Reliance Industries position. TCS receives the smallest allocation (10%), retained only for diversification given its low correlation with NIFTY 50 (0.4794) and the other stocks, despite its weak standalone performance. For a five-year horizon, this allocation is reasonable because it is weighted toward the stock and sector combination with the strongest historical risk-adjusted performance, while retaining exposure to two other stocks for diversification; however, no guarantee can be made that this pattern will persist over the coming five years.

12. Final Investment Recommendation

Reliance Industries is the best-performing individual stock on both an absolute basis (13.49% annualized return) and a risk-adjusted basis (Sharpe 0.3711; Treynor 6.49%), making it the strongest holding among the three. HDFC Bank is the most defensive, offering the lowest risk (16.97%) and a modest positive risk-adjusted return, while TCS is the weakest on every risk-adjusted measure, recording negative Sharpe and Treynor Ratios.

At the portfolio level, Portfolio B (40% HDFC Bank, 10% TCS, 50% Reliance Industries) is preferred over the equal-weighted Portfolio A, achieving a higher return (10.59% vs 8.38%) and a substantially higher Sharpe Ratio (0.2995 vs 0.1622) for only a marginal increase in risk (15.33% vs 14.66%) and beta (1.0166 vs 0.9550). Portfolio B is therefore recommended as the preferred structure for a moderately risk-averse investor.

Consistent with this, the ₹10,00,000 investment is allocated as ₹4,00,000 (40%) to HDFC Bank, ₹1,00,000 (10%) to TCS and ₹5,00,000 (50%) to Reliance Industries — concentrating capital in the stock with the strongest historical risk-adjusted performance, balanced by a defensive holding, while retaining a small diversifying position in TCS. Given the five-year horizon, this structure suits an investor willing to accept a modest increase in risk for a meaningfully improved risk-adjusted return, while maintaining exposure across three distinct large-cap equities rather than a single stock. This recommendation is based entirely on historical risk-return evidence and does not constitute a guarantee of future performance.

13. Limitations of the Analysis

- Historical returns, risk, beta and correlation are calculated from past price data and do not guarantee similar performance over the five-year horizon being planned for.
- The analysis is limited to three stocks, restricting the diversification benefit available compared to a broader portfolio.
- Results depend on the accuracy of the historical monthly price data used, and are specific to the January 2021–December 2025 study period; a different window could produce different rankings, particularly for TCS.
- Market conditions, company fundamentals and macroeconomic factors can change materially going forward.
- The 6% risk-free rate and 15% market-return assumption used in CAPM are simplifying inputs; monthly aggregation may also smooth over short-term volatility spikes, and CAPM itself relies on assumptions such as market efficiency and a linear risk-return relationship.

14. Conclusion

This report has evaluated the risk-return characteristics of HDFC Bank, TCS and Reliance Industries against the NIFTY 50 benchmark using monthly price data from January 2021 to December 2025, cross-validated in Excel and Python. Return and risk are not always proportionally related: Reliance Industries combines the highest risk with the highest return, while TCS combines high risk with the lowest return, underscoring the importance of risk-adjusted measures over raw return figures alone.

Reliance Industries is the best risk-adjusted performer, leading on both the Sharpe Ratio (0.3711) and Treynor Ratio (6.49%), while TCS is the weakest, with negative values on both measures. At the portfolio level, Portfolio B (40% HDFC Bank, 10% TCS, 50% Reliance Industries) outperformed the equal-weighted Portfolio A on a risk-adjusted basis, delivering a materially higher Sharpe Ratio (0.2995 vs 0.1622) for only a small increase in risk.

The Excel-versus-Python validation confirmed that core return, risk, beta and correlation calculations are consistent across platforms, while CAPM shows small differences for HDFC Bank and TCS (0.32 pp and 0.56 pp) attributable to formula-reference implementation within Excel, given that both platforms share the same 6% risk-free rate and 15% market-return assumption; for Reliance Industries, Excel and Python agree exactly at 16.39%.

On this evidence, the report recommends Portfolio B for a moderately risk-averse investor and proposes a ₹10,00,000 split of ₹4,00,000 to HDFC Bank, ₹1,00,000 to TCS and ₹5,00,000 to Reliance Industries for a five-year horizon. This directly reflects the objectives set out at the start of the report — measuring return, risk, beta, correlation, CAPM, Sharpe and Treynor for each stock, constructing and comparing two portfolios, validating Excel against Python, and translating the findings into a concrete investment allocation — while acknowledging that historical performance does not guarantee future results.